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Presentation on

# RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation

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# ABSTRACT

**1. Problem Statement**
Polysemous words can have multiple meanings, making it difficult for NLP systems to correctly understand text context and generate suitable titles.

**2. Objective**
To improve contextual understanding by resolving polysemous words and automatically generating meaningful, context-appropriate titles from textual content.

**3. Proposed Solution**
A RAG-enhanced Transformer framework is proposed to retrieve relevant external knowledge, disambiguate polysemous words, and generate suitable titles based on the given content

**4. Methodology**
The system preprocesses the input text, identifies ambiguous words, retrieves relevant context using RAG, performs contextual analysis using a Transformer model, and generates and evaluates an appropriate title.

**5. Expected Outcome**
The system is expected to provide better contextual understanding, accurate polysemy disambiguation, and relevant, meaningful titles for diverse textual content.

## Keywords:
-> RAG (Retrieval-Augmented Generation),
-> Transformer
-> NLP (Natural Language Processing),
-> Polysemy, Contextual Understanding,Disambiguation

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# INTRODUCTION

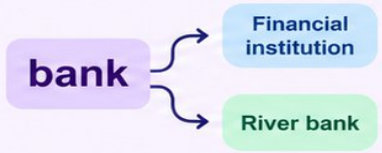
## KEY CONCEPTS

### 1 Natural Language Processing



Natural Language Processing (NLP) enables computers to **understand and interpret human language**, but understanding the correct meaning of words depends strongly on context.

### 2 Polysemy and Contextual Ambiguity




Polysemy occurs when a single word has **multiple related meanings**, making accurate interpretation difficult when the available context is limited or ambiguous.

### 3 Transformer-Based Understanding



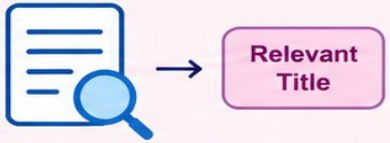

Transformer models provide **strong contextual understanding** by analysing relationships between words, but may require additional information when the input does not provide enough context.

### 4 RAG-Enhanced Context




Retrieval-Augmented Generation (RAG) **retrieves relevant information** from external knowledge sources and provides additional context to help the Transformer disambiguate polysemous words.

### 5 Automatic Title Generation




The proposed framework combines **RAG and Transformer-based analysis** to understand the complete content, resolve ambiguous meanings, and automatically generate a relevant and meaningful title.

# LITERATURE SURVEY

| S.No. | Author(s) & Year | Research / Method Used | Key Findings | Limitations / Research Gap | Proposed Enhancement |
|---|---|---|---|---|---|
| 1 | Bevilacqua et. al. (2021) | Word Sense Disambiguation (WSD) Survey | Reviewed recent advances and representation-learning improvements. | WSD still depends on available sense inventories and contextual resources. | Use external retrieved knowledge to provide richer context. |
| 2 | Li et al. (2024) | Transformer + BiLSTM for WSD | Combined Transformer with BiLSTM for Chinee WSb | Focused mainly on Chinese and specific datasets. | Extend disambiguation across diverse domains. |
| 3 | Abeysiriwardana & Sumanathilaka (2024) | Lexical Ambiguity Detection and WSD | Reviewed modern approaches for resolving ambiguity. | Scarcity of sense-annotated data and complex texts. | Integrate RAG and broader knowledge sources. |
| 4 | Yae et al. (2024/2025) | Large Language Models for WSD | Evaluated LLMs for word sense identification. | LLM performance varies and struggles with ambiguities. | Combine retrieval with LLM reasoning. |
| 5 | Zhang et al. (2025) | RoDEval – LLM-based WSD Evaluation | Evaluated LLMs' robustness and disambiguation | LLMs may lack sense coverage and show bias. | Enhance WSD with reliable external knowledge. |

| Software Requirements | Hardware Requirements |
|---|---|
| **Operating System:** Windows | **Processor:** Intel Core i5 |
| **Programming Language:** Python | **RAM:** Minimum 8 GB |
| **Development Environment:** VS Code | **Storage:** Minimum 20 GB available space |
| **NLP & Transformer:** Hugging Face Transformers | **GPU:** NVIDIA GPU with at least 4 GB VRAM |
| **RAG Framework:** Lang Chain / LlamaIndex | **Internet Connection:** Required for external knowledge retrieval and development resources |
| **Libraries:** PyTorch, Pandas, NumPy | **Input/Output Devices:** Keyboard, Mouse and Standard Monitor |